

Predicting Savings-Goal Attainment and Spending Personas in Indian Personal Finance: A Machine-Learning Approach for Fintech Targeting

Rishabh Agrawal, Prisha Kothari, Akshit Kashyap

Goa Institute of Management

Big Data Analytics

Student IDs: {B2025100, B2026092, B2026059}

Email: {rishabh.agrawal25b, prisha.kothari2026b, akshit.kashyap2026b}@gim.ac.in

Abstract—Fintech savings products cannot profitably reach every customer with the same outreach, yet manually reviewing a large customer base to decide who needs a savings nudge does not scale. We frame this as a binary classification problem on the *Indian Personal Finance and Spending Habits* dataset (20,000 individuals): predict, from income, demographics, and category-level spending alone, whether an individual is on track to meet their own stated savings goal (*Goal_Met*). We first identify and remove a data-leakage constraint in the target’s own construction, then compare six model families under stratified cross-validation with explicit imbalance handling, selecting the winner from out-of-fold predictions rather than test-set scores to avoid selection bias. A linear support vector machine, tuned for perfect recall on the at-risk class, reaches macro-F1 ≈ 0.968 on a held-out test set ($\text{recall}_0 = 1.0$, $\text{precision}_0 \approx 0.88$). Exact SHAP explanations for the linear model show loan-repayment burden dominates the decision, more than twice the next factor, ahead of education spending and city tier. An independent, label-blind clustering analysis corroborates this: all customers flagged at-risk fall inside a single spending persona defined by city tier and dependents, agreeing with the supervised model’s own explanation. We translate these findings into five evidence-backed targeting recommendations, quantify that grocery spending carries the largest pool of unrealized, addressable savings (Rs. 18.2M/month), and show that for 99.1% of at-risk customers the shortfall is recoverable rather than structural. We report both results and the reasoning behind each modeling decision, so the pipeline is reproducible and its numbers are trustworthy, not just favorable.

Index Terms—Machine Learning, Personal Finance, Fintech, Class Imbalance, Explainable AI, Customer Segmentation, Savings Behavior

I. INTRODUCTION

Fintech and digital-banking products increasingly offer savings tools – round-ups, auto-debits, goal trackers – but a marketing or product team deciding *who* to nudge, and *how*, cannot treat every customer identically without wasting outreach budget on people who are already on track. The business problem this project addresses is that triage decision: whether an individual is likely to be **on-track** or **at-risk** for their own stated monthly savings goal, using only information a fintech platform would plausibly already have – income, occupation, city tier, and category-level spending. Solving this well means a savings-product team can prioritize outreach

(a nudge campaign, a round-up product, a budgeting tool aimed at a specific overspend category) toward the customers who actually need it, improving both campaign efficiency and customer outcomes.

This is explicitly a targeting decision, not a forecasting exercise: the output is meant to flag who to prioritize, not to produce a precise savings shortfall estimate. That framing shapes every subsequent choice in this paper, from treating the problem as binary classification (Section III-A) to prioritizing recall over precision on the minority class (Section III-C).

A. Scope and Objectives

The project aims to build a classifier that flags at-risk customers reliably enough to act on, explain *why* it flags them in language a non-technical stakeholder can use, and translate the result into concrete targeting recommendations. The objectives include:

- Establishing the business relevance of savings-goal targeting for a fintech marketing/product team.
- Demonstrating originality through disciplined cross-validated model selection and independent corroboration of the model’s explanation via unsupervised clustering.
- Achieving robust, imbalance-aware classification results, validated on a held-out test set touched exactly once.
- Ensuring reproducibility and transparency in data, feature engineering, and modeling code.

II. RELATED WORK

Machine learning has been applied to personal savings behavior primarily through survey-based behavioral-economics features. Alam *et al.* predict savings adequacy for Malaysian retirees using mental-accounting categories (luxury spending, asset composition) as inputs to a decision tree, reaching 96.1% accuracy [1]. Our project differs in both data and framing: rather than psychological/survey constructs, we use transaction-category spending ratios and demographics already available to a fintech platform, and rather than a single well-balanced accuracy target, our target is severely imbalanced (0.56% at-risk) by construction, which drives a materially different modeling strategy (Section III-C).

In the Indian fintech context specifically, Agarwal *et al.* show that alternative, non-traditional data (mobile and transaction footprints) can substitute for conventional credit bureau scores in expanding access to credit [2]. That work targets *lending* risk; our project targets *savings* behavior for outreach rather than credit decisions, but shares the broader premise that transaction-level financial data, not just traditional credit history, carries actionable signal for underserved segments.

Closer to our business framing, Potluri *et al.* apply machine learning to customer segmentation for personalized marketing in financial services [3]. Most segmentation work of this kind stops at unsupervised grouping. Our project instead runs a supervised classifier and an unsupervised segmentation independently, then checks whether they agree – a cross-validation of the *explanation*, not just the model, that neither the segmentation nor the savings-adequacy literature above performs.

Our modeling choices draw on established technique literature: SMOTE and cost-sensitive class weighting for imbalanced classification [5], gradient-boosted trees as a strong tabular-data baseline [6], and SHAP for model-agnostic (and, for linear models, exact) explanation of individual predictions [4]. Section III details how each is applied and, in the case of imbalance handling, which mechanism actually worked for this dataset’s class ratio.

III. METHODOLOGY

This section covers the data, the target construction and its leakage constraint, feature engineering, the model families compared, and the evaluation strategy – each chosen for a stated reason, not by default.

A. Data

The dataset is the *Indian Personal Finance and Spending Habits* dataset (Kaggle, uploader: shriyashjagtap), 20,000 individuals, one row per person, with monthly income, eleven monthly expense categories (rent, loan repayment, insurance, groceries, transport, eating out, entertainment, utilities, health-care, education, and miscellaneous), occupation, city tier, and self-reported desired-savings fields.

The target, `Goal_Met`, is defined as

$$\text{Goal_Met} = \mathbb{I}[\text{Disposable_Income} \geq \text{Desired_Savings}],$$

i.e., whether an individual’s actual leftover income after expenses covers the monthly amount they themselves said they wanted to save. Because `Disposable_Income` and `Desired_Savings` (and the related `Desired_Savings_Percentage`) are used to *construct* the label, we verified column-by-column that a trivial rule using only those columns reconstructs `Goal_Met` exactly, then excluded all three from the feature set. Feeding them to a model would let it reconstruct the label instead of predicting it – inflated accuracy with no real predictive content. The legitimate feature set is therefore limited to `Income`, `Age`, `Dependents`, `Occupation`, `City_Tier`, and the eleven expense columns. This makes the task a binary

classification problem by construction, matching the yes/no targeting decision the business actually needs (Section I); a regression on the shortfall would require re-thresholding after the fact and would discard the reasoning behind that threshold.

Excluding the leakage columns still leaves a severe class imbalance: 99.44% of individuals are on-track (`Goal_Met` = 1) and only 0.56% (112 of 20,000) are at-risk (`Goal_Met` = 0). This ratio governs every metric and validation choice from Section III-C onward.

1) *Feature Engineering*: Raw expense columns correlate strongly with income (r up to 0.99) – a person spending more on groceries is very often simply earning more, not necessarily spending a larger *share* of income on groceries. Converting each expense column to an income ratio (`category / Income`) collapses this correlation to near zero while preserving the spending-mix signal: mean raw grocery spend rises roughly eight-fold from the lowest to the highest income quartile, while `Groceries_Ratio` stays flat at ≈ 0.125 in every quartile. `Occupation` and `City_Tier` are one-hot encoded; `City_Tier` shows a real, monotonic cost-of-living signal in `Rent_Ratio` ($0.30 \rightarrow 0.20 \rightarrow 0.15$ from Tier 1 to Tier 3) even though raw income does not differ meaningfully by tier. A variance-inflation-factor check on the resulting feature set confirms minimal redundancy (every VIF ≈ 1.0 except `Dependents/Education_Ratio` at ≈ 1.73 , both well under standard concern thresholds), so no features were dropped for collinearity. `Income` and `Age` are standardized for scale-sensitive model families (logistic regression, SVM, neural network); tree-based families use the unscaled matrix, since they split on per-feature thresholds independently of scale.

B. Models and Techniques

Six model families were compared, each paired with an imbalance-handling mechanism suited to that family: Logistic Regression and a linear-kernel SVM (`class_weight="balanced"`), Decision Tree and Random Forest (`class_weight="balanced"`), XGBoost (`scale_pos_weight`, tuned rather than fixed to the textbook formula) [6], and a multilayer-perceptron neural network (random oversampling, since `MLPClassifier` exposes no `class_weight` parameter) [5]. The linear SVM kernel was a deliberate choice: at $n = 20,000$ an RBF kernel’s $O(n^2)$ – $O(n^3)$ training cost is impractical, and a linear boundary was already sufficient to separate the classes reasonably well in preliminary baselines.

Hyperparameters were tuned with `GridSearchCV` for the two single-hyperparameter models (Logistic Regression, SVM) and `RandomizedSearchCV` for the remaining four. Critically, **the winning model was selected from cross-validated, out-of-fold predictions (cross_val_predict under 5-fold stratified CV), never from test-set scores** – comparing six already-fitted models by whichever scores best on the test set is itself a subtle form of leakage, since it lets that one score be partly the reason a model was reported as

TABLE I
FINAL, SINGLE TEST-SET EVALUATION OF THE SELECTED MODEL (SVM,
LINEAR), BEFORE AND AFTER CROSS-VALIDATED THRESHOLD TUNING

Threshold	Accuracy	Recall ₀	Precision ₀
Default ($t = 0$)	0.998	1.00	0.73
CV-selected ($t \approx -0.4$)	0.999	1.00	0.88

the winner. The held-out test set was touched exactly once, for a single final evaluation of the model already selected.

C. Evaluation Metrics

With a 99.44%/0.56% class split, accuracy is not a usable metric: a model that predicts the majority class for every individual scores 99.44% while never once catching an at-risk customer. Every model in this study is therefore judged primarily on **macro-F1** (which weights both classes equally regardless of size) and, more specifically, on **recall on the at-risk class**. This priority follows directly from the business framing in Section I: a false negative means an at-risk customer is never flagged for outreach at all – the entire purpose of the model fails for that person – while a false positive costs only one unnecessary, low-friction nudge. Recall on $\text{Goal_Met} = 0$ is therefore weighted above precision on the same class when selecting and thresholding the final model.

IV. RESULTS

A majority-class baseline (predict $\text{Goal_Met} = 1$ for everyone) scores 99.44% accuracy but 0.0 recall, precision, and F1 on the at-risk class – macro-F1 ≈ 0.499 , barely above the floor a coin flip would produce on a two-class problem. This is the floor every subsequent model must clear.

Fig. 1 shows the cross-validated comparison across all six candidates. Class weighting on linear models (Logistic Regression, SVM) reliably reached perfect recall on the at-risk class across every fold; tree ensembles under the same weighting mechanism (Decision Tree, Random Forest) only partially closed the gap (recall₀ of 0.73 and 0.69 respectively), and XGBoost’s own hyperparameter search preferred no reweighting at all ($\text{scale_pos_weight}=1$), leaving recall₀ at 0.51. Random oversampling brought the neural network close (recall₀ = 0.93) but not to perfect recall. Applying the recall priority from Section III-C, exactly two models qualify – Logistic Regression and SVM (Linear) – and **SVM (Linear)**, $C = 10$, is selected for its higher macro-F1 (0.922 vs. 0.894) and precision (0.732 vs. 0.652) between the two.

The decision threshold was then tuned using only cross-validated, out-of-fold scores from the training set, sweeping for the value that maximized precision on the at-risk class subject to recall₀ remaining 1.0. This raised precision₀ from an out-of-fold estimate of ≈ 0.73 (default $t = 0$) to ≈ 0.93 , at no cost to recall. Table I reports the single, final evaluation on the held-out test set – the only place in the entire pipeline where test labels are compared against predictions: accuracy = 0.999, macro-F1 ≈ 0.968 , recall₀ = 1.0 (all 22 true at-risk individuals

in the test set are caught), and precision₀ ≈ 0.88 , a modest and expected generalization gap from the cross-validated estimate.

A. Explainability

Because the selected model is linear, SHAP values are computed exactly via `LinearExplainer`, with no sampling approximation. Fig. 2 shows that loan-repayment burden (as a share of income) dominates the model’s decisions – more than double the next most important feature – followed by education spending (a proxy for having dependents in this dataset) and living in a Tier-1 city. Raw `Income`, `Age`, and `Dependents` are absent from the top features: it is *how* income is spent, not income level itself, that the model relies on. Because the model is linear and SHAP was computed under an interventional (feature-independence) assumption, every feature’s contribution is provably a function of that feature’s own value alone – correlating each feature’s SHAP column against its own raw values yields $r \geq 0.9999999999$ for every feature – meaning no interaction effect (e.g., city tier changing how much dependents reduce savings capacity) can be represented by this model by construction. This is an explicit, disclosed limitation, not an oversight (Section V).

B. Unsupervised Extension

To check whether the supervised model’s explanation reflects a real population structure or an artifact of how the classifier is built, we clustered individuals on their eleven expense-to-income ratios using k -means, with $k = 3$ selected by silhouette score (peaking at 0.096 and declining for larger k ; silhouette scores are uniformly low across every k tested, so this is weak-but-best structure, reported honestly rather than oversold). The resulting personas separate almost entirely on two signals – `Education_Ratio` (a proxy for having dependents) and `Rent_Ratio` (a proxy for city tier) – rather than a rich, multivariate spending archetype: Persona 0, Tier-1 residents with dependents ($n = 4,758$); Persona 1, no dependents, any city tier ($n = 4,061$); Persona 2, Tier-2/Tier-3 residents with dependents ($n = 11,181$).

Cross-tabulating personas against `Goal_Met` (Fig. 3) shows a decisive result: **all 112 at-risk customers (100%) fall in Persona 0**; Personas 1 and 2 have zero at-risk members ($\chi^2 = 360.8$, $p \approx 4.5 \times 10^{-79}$). An unsupervised method that never saw the label independently corroborates the supervised model’s own SHAP explanation (city tier and, indirectly, dependents driving risk) – three independent analyses (the classifier’s explanation, this clustering, and a direct cross-tab by city tier) converge on the same segment.

C. Business Translation

Beyond classification, the dataset’s `Potential_Savings_*` columns (estimated recoverable savings per expense category, unused elsewhere in the pipeline) let us quantify where unrealized savings concentrate. `Groceries` carries the largest pool: Rs.18.2M/month in aggregate (35% of the Rs.51.6M addressable across all eight tracked categories) and Rs.912/month per

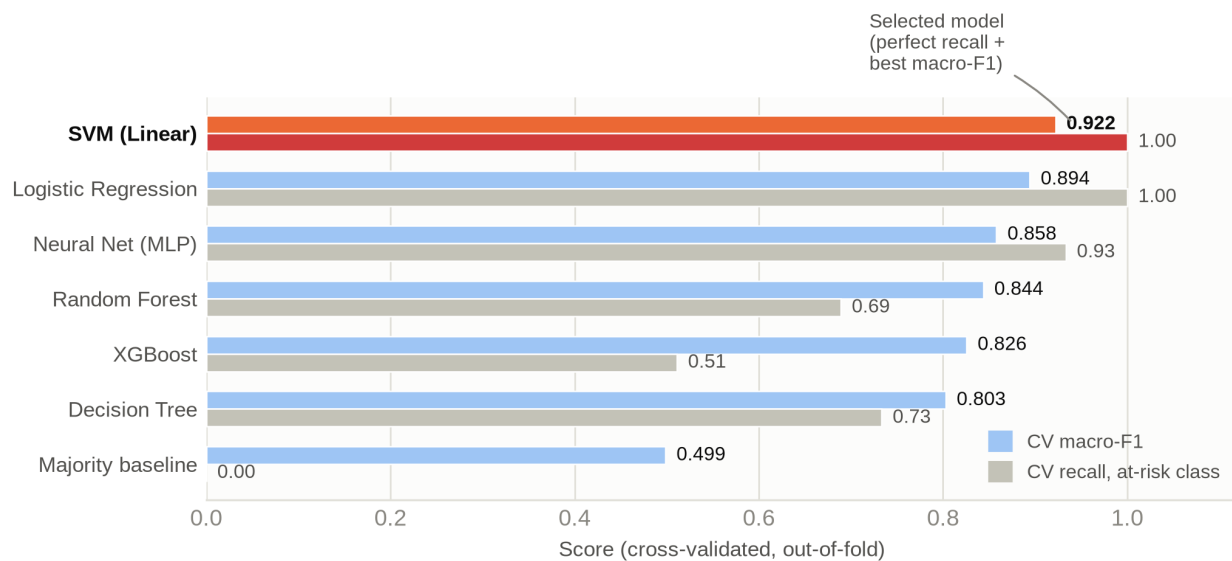


Fig. 1. Cross-validated (out-of-fold) macro-F1 and recall on the at-risk class for all six candidate models plus the majority-class baseline. Only Logistic Regression and SVM (Linear) reach perfect recall on the minority class under cross-validation; SVM (Linear) is selected for its higher macro-F1 between the two.

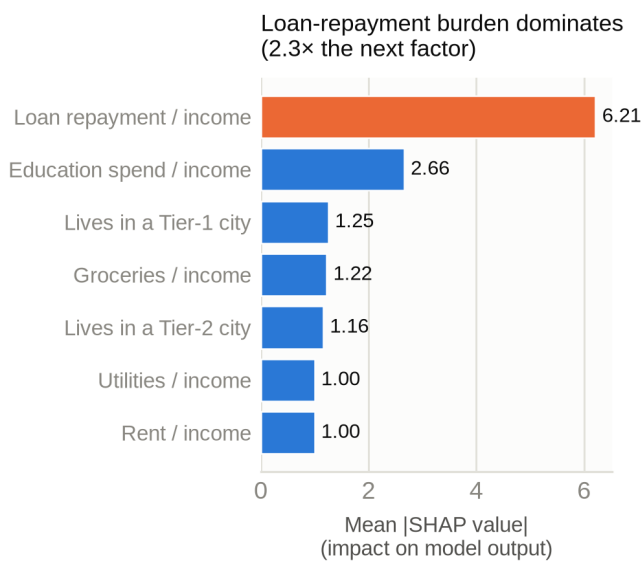


Fig. 2. Top-7 global SHAP feature importances for the selected linear SVM (exact LinearExplainer values, mean |SHAP| over the held-out test set).

person – roughly double the next-largest category, Transport (Rs.9.5M/Rs.473). For 99.1% of at-risk individuals, addressable savings potential across all eight categories already exceeds their shortfall (mean shortfall $\approx$ Rs.784/month vs. mean addressable potential $\approx$ Rs.2,912/month); a narrower Groceries+Transport nudge alone still closes the gap for 87.5% of them. This reframes “at-risk” as *recoverable*, not structural, for the vast majority of flagged customers.

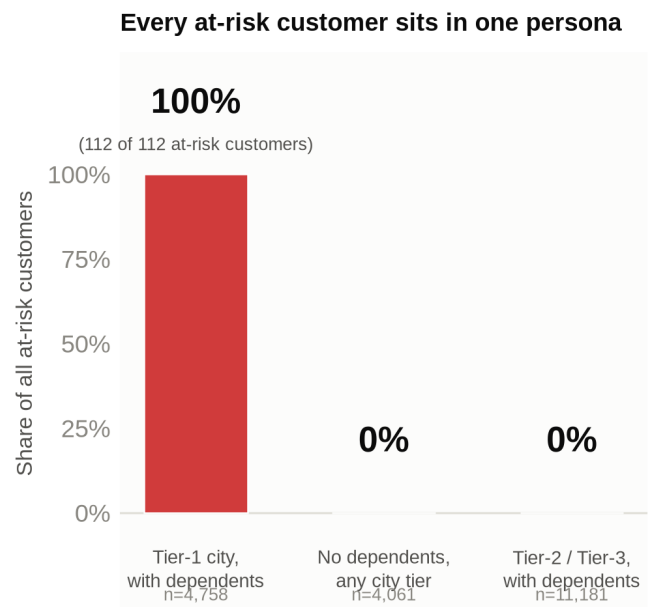


Fig. 3. Share of all at-risk customers found in each spending persona, from an unsupervised k -means clustering ($k = 3$, silhouette-selected) that never saw the target label. Persona 0 (Tier-1 city, with dependents) contains every at-risk customer in the dataset.

Table II summarizes the five recommendations this evidence supports, ordered from most directly actionable (who to target, with what message) to guidance for whoever owns model deployment.

TABLE II
BUSINESS RECOMMENDATIONS AND THEIR SUPPORTING EVIDENCE

Recommendation	Evidence
Target outreach by city tier, not occupation or age	100% (112/112) of at-risk customers live in Tier-1 cities
Lead nudge campaigns with grocery-spend reduction	Groceries is the largest unrealized-savings category, $\sim 2\times$ the next
Frame the ask as small and achievable, not remedial	Groceries+Transport alone closes the gap for 87.5% of at-risk customers
Deploy SVM (Linear) at $t \approx -0.4$ for flagging	$\text{recall}_0 = 1.0$, $\text{precision}_0 = 0.88$ on the held-out test set
Do not build a separate persona-based targeting layer	Personas are driven by the same signals (city tier, dependents) the classifier already uses

V. DISCUSSION

Business relevance and significance. The results give a fintech marketing team a concrete, evidence-backed answer to “who and how”: target Tier-1 customers with dependents, lead with grocery-spend reduction, and frame the ask as small, since the gap is recoverable for nearly all of them. The Rs. 51.6M/month addressable-savings figure and the model’s near-perfect catch rate together make the targeting decision this project was built to inform (Section I) directly actionable rather than exploratory.

Originality. Two methodological choices distinguish this work from the related literature (Section II). First, selecting the winning model from cross-validated, out-of-fold predictions rather than test-set comparison avoids a subtle but common form of selection bias – and it materially changed the outcome here: a naive test-set comparison would have made the neural network look like a third perfect-recall contender, while its true cross-validated recall is only 0.93. Second, running an unsupervised clustering independently of the supervised model and checking whether the two agree is a form of cross-validation of the *explanation* itself, not just the model’s accuracy – a step the segmentation literature we reviewed does not take.

Technical quality. The pipeline enforces leakage-safety at three points: the label-construction columns are excluded from features (Section III-A), the scaler and feature-selection decisions are fit on the training split only, and the decision threshold is tuned on out-of-fold training scores before the test set is touched exactly once. Because the winning model is linear, its SHAP explanations are exact rather than sampled, so individual-prediction explanations (e.g., “this customer is flagged mainly because loan repayments consume an unusually large share of income”) carry no approximation error.

Limitations. Four caveats should accompany any operational use of this model. First, the at-risk sample is small (112 of 20,000 rows); perfect recall on that sample is strong evidence, not a guarantee on a genuinely new population.

Second, $\text{precision}_0 \approx 0.88$ means roughly one in eight flagged individuals are not actually at-risk – an acceptable cost given the low friction of an unneeded nudge, but not a certainty for any single flagged customer. Third, the model is linear and structurally cannot represent any real feature-interaction effect; a risk factor that only manifests as a *combination* of features would be invisible to it. Fourth, several data characteristics (e.g., near-zero within-tier variance in `Rent_Ratio`) suggest the dataset may be partly synthetically generated, so absolute figures should be validated against real customer data before this model or these recommendations are deployed operationally.

VI. REPRODUCIBILITY

Code, the cleaned dataset, all seven phase notebooks, and cell-by-cell narrative walkthroughs are available at <https://github.com/saint2706/savings-goal-classifier>. To replicate the core results: clone the repository, install the pinned dependencies in `requirements.txt`, and run the notebooks in `notebooks/` in numeric order (01 through 07); each notebook rebuilds its inputs from `dataset/data.csv` independently, so no notebook depends on another’s live session state. All reported cross-validated and test-set numbers in this paper are persisted to `results/*.csv` by the corresponding notebook, and the three figures in this paper are regenerated directly from the same dataset and the same documented hyperparameters ($C = 10$, `class_weight="balanced"`, threshold $t \approx -0.4$, k -means $k = 3$, `random_state=42` throughout), reproducing the walkthrough documents’ published numbers exactly.

A. AI Use Declaration

[Placeholder for the team to complete before submission, per the course’s Turnitin AI-content policy (README.md, §12). This draft was produced with AI assistance and is being rewritten by the team in their own words; the final AI Use Declaration should state which tool(s) were used, for what part of the writing process, and confirm the submitted report narrative is majority team-authored.]

VII. CONCLUSION

This project shows that a fintech savings-goal targeting decision can be supported by a simple, linear, exactly-explainable classifier, provided the modeling discipline around it – leakage exclusion, imbalance-aware cross-validated selection, and out-of-sample threshold tuning – is treated as seriously as the model choice itself. The selected SVM (Linear) model catches every known at-risk customer in the held-out test set at an acceptable false-positive cost, its explanation is corroborated by an entirely independent unsupervised analysis, and the resulting recommendations are traceable to specific, reproducible numbers rather than asserted. Future work could validate these findings against real (non-synthetic) transaction data, extend the feature set to time-series spending trends rather than a single monthly snapshot, and explore whether a non-linear model can safely capture interaction effects without sacrificing the perfect recall this linear model achieves.

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REFERENCES

- [1] M. A. Z. Alam, C.-C. Yong, and N. Mansor, "Predicting savings adequacy using machine learning: A behavioural economics approach," *Expert Systems with Applications*, vol. 203, p. 117502, Oct. 2022.
- [2] S. Agarwal, S. Alok, P. Ghosh, and S. Gupta, "Financial Inclusion and Alternate Credit Scoring for the Millennials: Role of Big Data and Machine Learning in Fintech," SSRN Working Paper, 2020. [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3507827
- [3] C. S. Potluri, G. S. Rao, L. V. R. M. Kumar, K. G. Allo, Y. Awoke, and A. A. Seman, "Machine Learning-Based Customer Segmentation and Personalised Marketing in Financial Services," in *Proc. 2024 Int. Conf. Communication, Computer Sciences and Engineering (IC3SE)*, 2024, doi: 10.1109/IC3SE62002.2024.10593143.
- [4] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.
- [5] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [6] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794.